

VTASK 2: STOCK PORTFOLIO TRACKER

This project is a simple Python-based Stock Portfolio Tracker. It allows users to enter stock names and quantities, calculates the total investment value using predefined stock prices, and optionally saves the portfolio details into a text or CSV file.

1. PROJECT OBJECTIVE

Build a simple stock portfolio tracker that calculates the total investment value based on manually defined stock prices.

2. PROJECT FEATURES

- User can enter stock names and quantity.
 - Uses a predefined dictionary for stock prices.
 - Calculates total investment value automatically.
 - Displays portfolio summary clearly.
 - Optional file saving using TXT or CSV format.
 - Beginner-friendly Python project.
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3. TECHNOLOGIES USED

- Python 3
 - Dictionary Data Structure
 - Input / Output Operations
 - Loops and Conditional Statements
 - File Handling (TXT / CSV)
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4. KEY CONCEPTS USED

- Dictionary
 - User Input
 - Arithmetic Operations
 - File Handling
 - Loops and Conditions
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5. SAMPLE STOCK PRICE DICTIONARY

```
stock_prices = {  
    'AAPL': 188,  
    'TSLA': 258,
```

```
'GOOGL': 135,  
'MSFT': 330  
}
```

6. HOW THE PROGRAM WORKS

Step 1: User enters stock name.
Step 2: User enters quantity of shares.
Step 3: Program checks stock price from dictionary.
Step 4: Total investment value is calculated.
Step 5: Portfolio summary is displayed.
Step 6: Data can optionally be saved into TXT or CSV file.

7. SAMPLE OUTPUT

Enter stock name: AAPL
Enter quantity: 5

Enter stock name: TSLA
Enter quantity: 2

Portfolio Summary:
AAPL = $5 \times 188 = 940$
TSLA = $2 \times 258 = 516$

Total Investment Value = 1456

8. PROJECT STRUCTURE

```
stock_portfolio_tracker/  
■■■■ stock_tracker.py  
■■■■ README.md  
■■■■ portfolio.txt  
■■■■ portfolio.csv
```

9. FUTURE IMPROVEMENTS

- Add real-time stock price API integration.
 - Create graphical user interface (GUI).
 - Add profit/loss calculation.
 - Store portfolio permanently using databases.
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10. INTERNSHIP DETAILS

Internship Domain: Python Programming
Project Name: Stock Portfolio Tracker
Organization: CodeAlpha

11. CONCLUSION

This project demonstrates basic Python programming concepts such as dictionaries, user input, arithmetic calculations, and file handling. It is a beginner-friendly project useful for understanding simple financial tracking systems.
